

PROJECT ABSTRACT

ReliefLink – Flood Relief Item Availability Portal

1. Problem Statement (Existing System)

During natural disasters such as floods, landslides, earthquakes, cyclones, heatwaves, and fires, affected people often struggle to access timely emergency assistance, relief camps, volunteers, and essential relief supplies. Existing disaster management systems mainly provide emergency contacts or weather updates but lack a centralized platform that connects victims, volunteers, NGOs, relief centers, and administrators. Information is often shared through phone calls, social media, and messaging applications, making it difficult to verify and coordinate rescue operations. The absence of real-time SOS tracking, volunteer coordination, inventory management, disaster alerts, and relief request monitoring leads to delayed response, inefficient resource distribution, and poor disaster management.

2. Objective

The objective of **ReliefLink** is to develop a centralized web-based Disaster Relief and Emergency Coordination Platform that enables affected people, volunteers, NGOs, relief centers, and administrators to collaborate during emergencies. The system allows users to send SOS requests, request relief materials, locate nearby relief camps and emergency services, receive disaster alerts, monitor weather conditions, and track relief requests in real time. It aims to improve disaster response, enhance coordination among stakeholders, ensure transparent resource distribution, and provide faster emergency assistance during disaster situations.

3. Proposed Solution

ReliefLink is a full-stack web application designed to support disaster management during floods, landslides, earthquakes, cyclones, heatwaves, fires, and other natural disasters.

Affected persons can quickly access the platform using **phone number verification**, allowing faster emergency access during disaster situations. Users can send SOS requests, update their "**I'm Safe**" status, locate nearby relief camps and emergency services, request essential relief items such as food, drinking water, medicines, clothes, sanitary products, baby care essentials, blankets, and hygiene kits, track request status, and receive emergency notifications.

Volunteers can register, specify their skills, view nearby SOS and relief requests, accept rescue and delivery tasks, navigate safely using interactive maps, deliver relief materials, verify completed tasks, and update task progress in real time.

NGOs and Relief Centers can create and manage relief camps, maintain inventories of essential supplies, receive donations, approve relief requests, allocate available resources, assign volunteers, and monitor relief distribution through a centralized dashboard.

Administrators manage users, NGOs, volunteers, relief camps, SOS requests, disaster alerts, inventory, reports, and overall system activities. The platform also provides live weather information, disaster alerts, local and state disaster news, hazard broadcast zones, browser push notifications, interactive maps, and real-time updates to improve disaster awareness and emergency coordination.

4. Technology Stack

Component	Technology
Frontend	React.js + Vite
Backend	Node.js + Express.js
Database	MongoDB
Authentication	JWT (JSON Web Token)
Maps	Leaflet + OpenStreetMap
Real-time Communication	Socket.io
Weather API	OpenWeather API
Notifications	Website Push Notifications
Image Storage	Cloudinary (<i>Optional</i>)

5. Benefits of the Project

- Provides a centralized platform for disaster relief and emergency coordination.
- Enables quick SOS reporting and faster emergency response.
- Simplifies relief item requests and distribution.
- Helps NGOs manage inventories, donations, and relief camps efficiently.
- Improves coordination between victims, volunteers, NGOs, and administrators.
- Supports real-time weather updates, disaster alerts, and emergency notifications.
- Helps users locate nearby shelters, relief camps, and emergency services.
- Enhances transparency through request tracking and live status updates.
- Reduces response time and improves disaster management efficiency.

MODULE DESCRIPTION

Admin

- Dashboard
- Manage Users
- Manage NGOs
- Manage Volunteers
- Manage Relief Camps
- Manage Inventory
- Manage SOS Requests

- Manage Disaster Alerts
 - Broadcast Hazard Notifications
 - Reports & Analytics
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NGO / Relief Center

- Register / Login
 - Manage Relief Camps
 - Manage Inventory
 - Receive Donations
 - Approve Relief Requests
 - Allocate Resources
 - Assign Volunteers
 - Track Relief Distribution
 - View Reports
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Volunteer

- Register / Login
 - Register Skills
 - View Nearby SOS Requests
 - View Relief Tasks
 - Accept Rescue & Delivery Tasks
 - Deliver Relief Items
 - Verify Task Completion
 - Update Task Status
 - Safe Route Navigation
 - Activity Dashboard
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User (Affected Person)

- Quick Phone Number Login
 - Send SOS Request
 - Update "I'm Safe" Status
 - Request Relief Items
 - Find Nearby Relief Camps
 - View Nearby Emergency Services
 - View Emergency Contacts
 - Track Relief Request Status
 - Receive Push Notifications
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Main Modules

1. Authentication Module
2. Affected Person Module
3. Volunteer Module
4. NGO & Relief Center Module

5. SOS & Emergency Module
6. Relief Request & Donation Module
7. Inventory Management Module
8. Relief Camp Management Module
9. Maps & Navigation Module
10. Weather, News & Disaster Alert Module
11. Notification Module
12. Reports & Analytics
13. Admin Dashboard

Total Modules: 13

Relief Services

- Food Kits
- Drinking Water
- Medicines
- Clothes
- Sanitary Products
- Blankets & Shelter
- Hygiene Kits
- Ambulance Assistance
- Medical Assistance